

Problem 2

(a) Show $\oint_C (f \nabla f) \cdot d\vec{r} = 0$

⊛ Note: Mixed partials are equal since f has continuous 2nd partial derivatives

Consider

$$\begin{aligned}\text{curl}(f \nabla f) &= \text{curl}([ff_x, ff_y, ff_z]) \\ &= [(ff_z)_y - (ff_y)_z, (ff_x)_z - (ff_z)_x, (ff_y)_x - (ff_x)_y] \\ &= [f_y f_z + f f_{zy} - f_z f_y - f f_{yz}, f_z f_x + f f_{xz} - f_x f_z - f f_{zx}, \\ &\quad f_x f_y + f f_{yx} - f_y f_x - f f_{xy}]\end{aligned}$$

⊛

$$= [0, 0, 0]$$

by Stokes' Thm:

$$\oint_C (f \nabla f) \cdot d\vec{r} = \iint_S \text{curl}(\vec{F}) \cdot \vec{n} dA = 0$$

(b) Show $\oint_C f \nabla g + g \nabla f$

$$\text{curl}(f \nabla g + g \nabla f) = \text{curl}(\nabla(fg)) = 0$$

curl of any gradient is 0